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VOLUME 25
ISSUE 4

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Nutrition in Critical Care: Evidence-Based Management

Nutritional support is vital in critical care, yet optimal timing, dose, and composition are debated. Guidelines often conflict with recent trials showing that even lower early goals versus aggressive or early escalating dose may be harmful or not relevant. This review synthesises current evidence, highlighting guideline–trial discrepancies and proposing phase-specific strategies for ICU management.

Physiological Basis and Nutritional Risk Stratification in Critical Illness

Response to nutritional therapy in critically ill patients is influenced by profound metabolic heterogeneity, which explains why many nutrition trials have reported conflicting results. During the acute phase of critical illness, a dominant catabolic state develops, characterised by hypermetabolism, anabolic resistance and altered nutrient utilisation. In this context, the organism relies on endogenous substrate mobilisation rather than incorporation of exogenous nutrients. Transitioning from catabolism to anabolism, essential for tissue repair, depends on factors such as inflammation, organ function, hormonal balance, haemodynamic stability, and adequate oxygen delivery (Reignier et al. 2025b). On the other hand, catabolism/autophagy has been hypothesised as a favourable process in which tissue and cellular damage can be controlled, providing nutrients to continue tissue repair and organ function (Van Dyck et al. 2018).

Variability arises from baseline characteristics, including age, comorbidities, pre-existing nutritional status, frailty, and genetic polymorphisms that modulate energy expenditure and immune-inflammatory responses. At the time, there are no clear indicators of metabolic readiness to guide the timing and intensity of nutritional support. Such indicators underline their importance given that inappropriate nutrient dosing at a given time carries risks: overfeeding may worsen hyperglycaemia, hyperlipidaemia, and hepatic damage, whereas underfeeding accelerates muscle wasting, predisposes to infections, dependency and delays recovery.

The Nutrition Risk in Critically ill (NUTRIC) score is the first instrument developed to identify critically ill patients most likely to benefit from early and targeted nutrition therapy. Incorporating age, comorbidities, illness severity (APACHE II, SOFA), and inflammatory status (IL-6 in the original version), it stratifies nutritional risk, linking higher scores to greater mortality, longer ICU stay, and increased vulnerability to muscle wasting. Importantly, the score highlights that not all ICU patients require aggressive feeding; instead, nutrition should be prioritised in those with high NUTRIC scores (≥ 5), where the impact on outcomes is most significant. Controversy might arise given that this score relies mostly on inflammation/severity markers and lacks "nutrition" variables (Heyland et al. 2011).

The Global Leadership Initiative on Malnutrition (GLIM) criteria provide a complementary framework for diagnosing malnutrition, combining phenotypic (weight loss, low BMI, reduced muscle mass) and aetiologic (reduced intake, inflammation or disease burden) components, thus broadening the ability to identify patients who may derive the greatest benefit from timely and individualised nutrition interventions. Given the dynamic changes during critical illness, baseline status alone may not be an optimal guide for nutritional support. Thus, the physiological rationale for nutrition in critical care is not the application of uniform prescriptions, but rather the tailoring of therapy to the patient's metabolic trajectory. Tools such as indirect calorimetry, theoretically the most accurate method to estimate energy expenditure, body composition analysis, such as bioelectrical impedance, and dynamic biomarkers may help determine the optimal moment to shift from catabolic to anabolic support (Reignier et al. 2025b).

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Guideline Updates in Nutrition for the Critically Ill

International guidelines on nutritional support in critically ill patients, issued by the European Society for Clinical Nutrition and Metabolism (ESPEN) in 2023 and by the American Society for Parenteral and Enteral Nutrition/Society of Critical Care Medicine (ASPEN/SCCM) in 2022, were largely informed by physiological rationale, observational studies, and small heterogeneous trials. Regarding enteral nutrition (EN), both societies recommend early EN within 24–48 hours of ICU admission in patients with a functional gastrointestinal tract and haemodynamic stability (not universally defined at the time). ESPEN advises starting with $\leq 70\%$ of estimated energy requirements during the first week, then advancing to 20–25 kcal/kg/day, with a progressive protein target of ~ 1.3 g/kg/day. When indirect calorimetry is used, advancing to 80–100% of the measured energy expenditure after day 3 is recommended by ESPEN (Singer et al. 2023). ASPEN/SCCM recommend 12–25 kcal/kg/day and an aggressive approach of 1.2–2.0 g/kg/day of

protein during the first 7 to 10 days. Early parenteral nutrition (PN) is recommended by both societies in the context of high-risk or severe malnutrition with EN contraindications, while PN in low-risk and well-nourished patients could be delayed for one week. Calories and protein doses remain as mentioned by ESPEN; on the other hand, ASPEN suggest ≤ 20 Kcal/Kg and ≥ 1.3 g/Kg during the first week of PN (Compher et al. 2022).

These recommendations, which historically favoured early nutrition with relatively high calorie and protein provision, are now evolving in light of recent large trials with stricter methodologies and patient-centred outcomes, steering critical care nutrition toward a “less is more” approach. Based on these pragmatic randomised studies, the 2025 French Intensive Care

Society (SRLF), the Société Francophone de Nutrition Clinique et Métabolisme (SFNCM) and the French-Speaking Group of Pediatric Emergency Physicians and Intensivists (GFRUP) guideline update has challenged this paradigm (Reignier et al. 2025a). Current evidence supports lower early energy (6–8 kcal/kg/day) and protein (0.2–0.9 g/kg/day) targets, associated with fewer infections, shorter ICU length of stay, improved survival, and less harm in patients with acute kidney injury and multiple organ failure. This emerging evidence underscores the urgent need for international guideline updates and invites intensivists to adopt a phase-specific, outcome-oriented approach in the nutritional management of critically ill patients (Table 1).

		ASPEN 2016-2021			ESPEN 2023			SRLF/GFRUP 2025		
		When?	Calories	Protein	When?	Calories	Protein	When?	Calories	Protein
Early	EN	≤48h	12-25Kcal/Kg	≥1.2g/Kg	≤48h	Provide <70% of EE; advance to 80–100% when EE is measured by IC.	Escalate to 1.3g/Kg	≤48h	6-8Kcal/Kg IC probably wouldn't change outcomes	0.2-0.9g/Kg
	PN	If high risk or severely malnourished and EN not possible: ASAP	≤20Kcal/Kg		If high risk or severely malnourished and EN not possible: early and progressive	Within the first week				
	SPN	Rather optimise EN			Rather optimise EN			No benefit		
Late	EN		Increase to goal by IC or 25-30Kcal/Kg	1-2-2.0g/Kg if no NB		Increase to goal	1.3g/Kg		20-30Kcal/Kg	1.0-1.3g/Kg
	PN	If low risk or well-nourished and EN not possible			Every patient with EN contraindication within 3 to 7 days					
	SPN	Every patient not reaching >60% of the goals			Every patient with no full EN tolerance despite optimisation			No clear indication		

Table 1. Comparison of international guidelines for critical care nutrition. Abbreviations: ASPEN, American Society for Parenteral and Enteral Nutrition; ESPEN, European Society for Clinical Nutrition and Metabolism; SRLF, Société de Réanimation de Langue Française; GFRUP, Groupe Francophone de Réanimation et d'Urgences Pédiatriques; EN, enteral nutrition; PN, parenteral nutrition; SPN, supplemental parenteral nutrition; EE, energy expenditure; IC, indirect calorimetry; NB, nitrogen balance.

Trial	Population	Intervention	Comparator	Key Outcomes
NUTRIREA-3 (2023)	3,044 ICU patients with shock, mechanically ventilated (61 French ICUs)	Low feeding: ~6 kcal/kg/day + 0.2–0.4 g/kg/day protein	Guideline-based feeding: 25 kcal/kg/day + 1.0–1.3 g/kg/day protein	90-day mortality: no difference; earlier readiness for ICU discharge (8 vs 9 days); fewer complications (vomiting, diarrhoea, bowel ischaemia, liver dysfunction).
EFFORT Protein (2023)	1,301 nutritionally high-risk ICU patients (85 ICUs, 16 countries)	High protein: ≥2.2 g/kg/day	Usual protein: ≤1.2 g/kg/day	No difference in time to discharge alive at 60 days or 60-day mortality; possible harm in AKI or high SOFA subgroups.
PRECISE (2025)	935 ICU patients, mechanically ventilated	High protein: 2.0 g/kg/day	Standard protein: 1.3 g/kg/day	Bayesian analysis: 0% probability of benefit, ~47% probability of harm (mortality, QoL); no improvement in 60-day mortality.
TARGET (2025)	3,397 ICU patients in Australia & New Zealand (8 ICUs)	High-protein EN formula: 100 g/L	Standard EN formula: 63 g/L	No difference in days alive and out of hospital at 90 days; mortality, ICU/hospital LOS, ventilation duration all neutral.

Table 2. Comparative summary of major nutrition trials in the critically ill: NUTRIREA-3, EFFORT, PRECISE, and TARGET. Abbreviations: ICU, intensive care unit; MV, mechanical ventilation; EN, enteral nutrition; PN, parenteral nutrition; LOS, length of stay; AKI, acute kidney injury; SOFA, Sequential Organ Failure Assessment; QoL, quality of life; HR, hazard ratio; RR, risk ratio; CI, confidence interval.

Less is More: Contemporary Evidence on Nutrition for the Critically Ill

Recent large, randomised trials consistently suggest that in critically ill ICU patients, less may be more when it comes to early energy and protein delivery. These findings challenge long-standing dogma favouring early full-dose feeding and instead support a more restrained, individualised strategy aligned with metabolic tolerance (**Table 2**).

The NUTRIREA-3 trial (Reignier et al. 2023), the largest RCT to date on acute-phase nutrition, randomised 3,044 ventilated patients in shock across 61 French ICUs to low-calorie, low-protein feeding (~6 kcal/kg/day; 0.2–0.4 g/kg/day protein) versus targets reached within 7 days (25 kcal/kg/day; 1.0–1.3 g/kg/day protein). Ninety-day mortality was similar (41.3% vs. 42.8%; $p=0.41$), but patients in the low-intake group achieved earlier readiness for ICU discharge (median 8 vs. 9 days; HR 1.12; $p=0.015$) and experienced fewer gastrointestinal compli-

cations, including vomiting, diarrhoea, bowel ischaemia, and liver damage. Low target group showed less incidence of insulin administration and hypophosphatemia, with no differences in hypoglycaemia events. These results support the safety and potential advantages of permissive calorie and protein restriction during the first week of critical illness.

The effect of higher protein dosing in critically ill patients with high nutritional risk (EFFORT Protein Trial) examined protein dosing in 1,301 nutritionally high-risk ICU patients across 85 ICUs in 16 countries (Heyland et al. 2023). High protein intake (≥2.2 g/kg/day) showed no benefit over usual protein (≤1.2 g/kg/day) in time to discharge alive at 60 days (HR 0.91; $p=0.27$) or 60-day mortality (RR 1.08; $p=0.27$). Subgroup analyses suggested possible harm in patients with acute kidney injury or elevated SOFA scores, underscoring the need to adapt prescriptions to organ function and metabolic capacity. It's worth underlining that every patient in the EFFORT protein

trial was considered at least at nutritional risk, challenging the concept of individualised nutrition therapies according to baseline status.

The Effect of high versus standard protein provision on functional recovery in people with critical illness (PRECISE) trial reanalysed under a Bayesian framework (Schouteden et al. 2025), randomised 935 ventilated patients to high (2.0 g/kg/day) versus standard (1.3 g/kg/day) protein via EN. Bayesian modelling demonstrated a 0% probability of benefit and a 15% probability of clinically relevant harm for health-related quality of life (EQ-5D-5L). For 60-day mortality, the probability of benefit was only 8%, while the probability of clinically important harm exceeded 47%. Although a minor functional signal appeared in the 6-minute walk test, attrition bias (<31% completion) limited interpretation. Collectively, these findings reinforce that aggressive protein delivery may not improve outcomes and could plausibly increase harm.

Trial	Population & Design	Intervention	Comparator	Main Outcomes	Key Findings
REDOXS	1,223 critically ill patients with multiorgan failure, many with sepsis; multicentre RCT	High-dose glutamine (0.35 g/kg/day IV + 30 g/day enteral) ± antioxidants	Placebo	28-day mortality, organ failure	Glutamine associated with ↑ mortality (32.4% vs. 27.2%, HR 1.28; p=0.05), especially in renal/hepatic dysfunction; no clinical benefit.
MetaPlus	301 mechanically ventilated ICU patients expected to stay ≥72h; multicentre RCT	Immune-enhancing EN (enriched with glutamine, omega-3 FA, selenium, antioxidants)	Iso-caloric, isonitrogenous standard EN	Mortality, infections, ICU/hospital LOS	No difference in infections or LOS; ↑ mortality in medical ICU subgroup (54% vs. 35%; p=0.03).

Table 3. Summary of glutamine supplementation major trials in critical illness: REDOXS and MetaPlus. Abbreviations: ICU, intensive care unit; EN, enteral nutrition; RCT, randomised controlled trial; LOS, length of stay; HR, hazard ratio; NEJM, New England Journal of Medicine; FA, fatty acids.

The TARGET Protein Trial (Summers et al. 2025), a cluster-randomised crossover study in 8 ICUs in Australia and New Zealand, tested high-protein (100 g/L) versus usual-protein (63 g/L) isocaloric enteral formulas in 3,397 critically ill patients. The primary outcome, days alive and out of hospital at 90 days, showed no difference (median 62 vs. 64 days; adjusted median difference -1.97; 95% CI -7.24 to 3.30; p=0.46). Secondary outcomes, including mortality, ventilation duration, length of stay, tracheostomy, and renal replacement therapy, were also unaffected. This trial further reinforces the emerging narrative: augmenting enteral protein beyond standard levels during the acute phase provides no measurable benefit, strengthening the case for a cautious, individualised approach over aggressive supplementation.

The original Effect of Early Nutritional Support on Frailty, Functional Outcomes and Recovery of Malnourished Medical Inpatients Trial (EFFORT) (Schuetz et al. 2019) was carried out in non-critically ill patients with high nutritional risk at eight Swiss hospitals. In this trial, patients were randomised to

receive protocol intervention aimed to reach calorie and protein goals versus standard of care (1050 and 1038, respectively). Individualised approach showed less adverse clinical outcomes (OR 0.79, p=0.023) and mortality (OR 0.65, p=0.011), with no difference in side effects. External validity of such a trial is not possible given the target population (non-critically ill patients), but a secondary analysis dividing the total sample according to their CRP (C-Reactive protein) concentrations revealed that the benefit from target goals disappeared in those with CRP >100mg/L. This can help to rethink that people severely ill or in a state of high inflammation (such as critically ill) might not respond to an aggressive nutritional approach (Merker et al. 2020).

Glutamine and Arginine in Critical Illness

The use of glutamine supplementation in critically ill patients has been progressively abandoned following high-quality evidence from large, randomised trials. Studies such as REDuc-

ing Deaths due to OXidative Stress (REDOXS) (Heyland et al. 2013) and MetaPlus (Van Zanten et al. 2014) demonstrated not only the absence of clinical benefit but also an increased risk of harm, particularly higher mortality in patients with sepsis, multiorgan failure, or renal and hepatic dysfunction. In light of these findings, recent international guidelines consistently recommend against the routine use of glutamine in critically ill populations (Table 3).

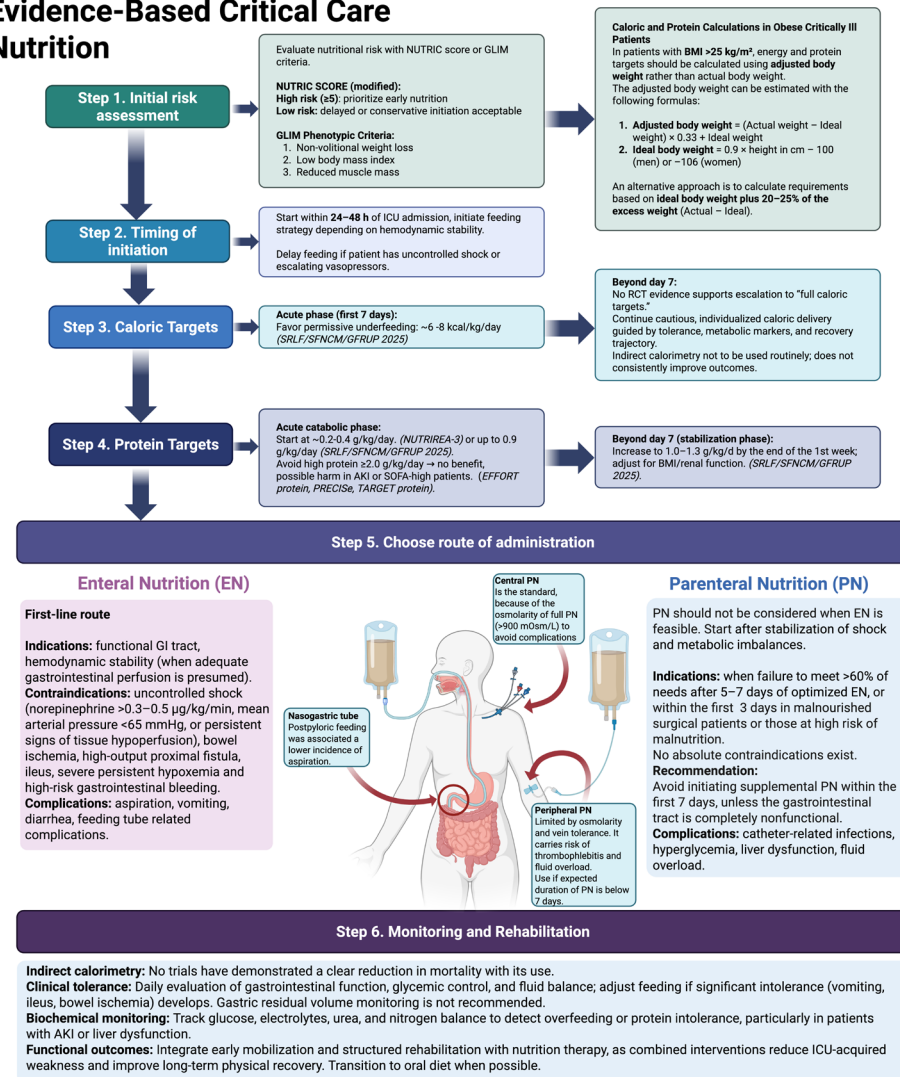
Similarly, arginine supplementation has raised concerns in the context of sepsis and shock due to its role as a nitric oxide precursor, with the potential to exacerbate vasoplegia and haemodynamic instability. While some perioperative surgical patients without sepsis might theoretically benefit from arginine's immunomodulatory properties, current evidence does not support its routine use in the ICU setting. Accordingly, the most recent guidelines advise against arginine supplementation in septic or haemodynamically unstable patients, reinforcing the broader shift away from aggressive immunonutrition toward safer, individualised nutritional support.

Beyond the ICU: Linking Nutrition to Long-Term Outcomes

The relevance of nutrition therapy extends beyond the acute ICU stay into the post-ICU period, where many patients develop post-intensive care syndrome (PICS), a constellation of physical, cognitive, and psychological impairments that may persist long after hospital discharge. Malnutrition and prolonged immobilisation are frequent among critically ill patients and strongly associated with poor outcomes, including accelerated muscle atrophy, impaired strength, and delayed weaning from mechanical ventilation. These factors exacerbate ICU-acquired weakness (ICU-AW), with muscle loss reaching up to 20% of total body mass within the first week of critical illness, directly impacting long-term mortality and quality of life (Yang et al. 2018).

To mitigate these risks, nutritional therapy should be conceived as a continuum that extends beyond ICU discharge, integrated with early mobilisation and structured rehabilitation

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programmes. Such a combined strategy has the potential to limit muscle wasting, preserve physical function, and reduce the burden of PICS. Nevertheless, there is still insufficient evidence to determine the optimal nutritional prescription for the post-ICU phase. Large, high-quality trials specifically addressing nutrition and rehabilitation with PICS as a primary endpoint are urgently needed to guide practice and improve long-term outcomes.

Cost Implications of Restrictive Nutrition in Critical Illness

Restrictive feeding strategies not only align with recent evidence on safety and clinical outcomes but may also reduce healthcare costs by lowering formula consumption. For example, for a 70-kg critically ill patient receiving nutrition over 7 days, standard targets of 20–25 kcal/kg/day equate to ~1,400–1,750 kcal/day, while a restrictive approach of 6–10 kcal/kg/day provides only ~420–700 kcal/day. This represents a daily reduction of ~980–1,330 kcal, or ~6,900–9,300 kcal over one week. Assuming a commercial enteral formula costs 2.5 USD per 1,000 kcal, the restrictive regimen would save approximately 17–23 USD per week per patient. Extrapolated to an ICU admitting 20 such patients per month, this translates into 4,080–5,520 USD in annual savings, underscoring both the clinical and economic implications of lower energy provision during the acute phase of critical illness.

Conclusion

Over the past decade, high-quality randomised trials and updated international guidelines have reshaped our understanding of nutrition in critical illness. Evidence consistently shows that aggressive early provision of calories and protein not only fails to improve outcomes but may also increase complications and mortality in selected subgroups. Strategies emphasising permissive underfeeding during the acute phase, gradual progression of protein delivery, and avoidance of immunonutrition supplements such as glutamine or arginine align better with the metabolic trajectory of critically ill patients (Figure 1).

Figure 1. Evidence-based critical care nutrition algorithm. Illustration created with BioRender.com. Abbreviations: BMI, body mass index; EN, enteral nutrition; PN, parenteral nutrition; MAP, mean arterial pressure; NE, norepinephrine; GI, gastrointestinal; ICU, intensive care unit; RCT, randomised controlled trial; SRLF, Société de Réanimation de Langue Française; SFNCM, Société Francophone de Nutrition Clinique et Métabolisme; GFRUP, Groupe Francophone de Réanimation et d'Urgences Pédiatriques; EFFORT Protein, Effect of Early Protein Feeding in Critically Ill Patients; PRECISE, Pragmatic Randomized Evaluation of Individualized Protein in Critical Illness; TARGET PROTEIN, Trial of Protein Dose in Critically Ill Patients; AKI, acute kidney injury; SOFA, Sequential Organ Failure Assessment

Looking ahead, nutrition therapy must be individualised, phase-specific, and closely integrated with rehabilitation to support both short- and long-term recovery. While current data support restraint in the acute ICU setting, uncertainties remain, particularly regarding optimal prescriptions for the

post-ICU phase and prevention of PICS. Future research should focus on tailoring nutrition to metabolic readiness, functional outcomes, and patient-centred endpoints, positioning nutrition not as a one-size-fits-all intervention but as a dynamic cornerstone of critical care.

Conflict of Interest

None.

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